

Appendix D

Transformer Data

Table D.1 Impedances for two-winding power transformers.

High-voltage winding insulation class kV	Low-voltage winding insulation class kV	Impedance limits in percent			
		Class OA OW OA/FA ^{a)} OA/FA/FOA ^{a)}		Class FOA FOW	
		Min	Max	Min	Max
15	15	4.5	7.0	6.75	10.5
25	15	5.5	8.0	8.25	12.0
34.5	15	6.0	8.0	9.0	12.0
	25	6.5	9.0	9.75	13.5
46	25	6.5	9.0	9.75	13.5
	34.5	7.0	10.0	10.5	15.0
69	34.5	7.0	10.0	10.5	15.0
	46	8.0	11.0	12.0	16.5
92	34.5	7.5	10.5	11.25	15.75
	69	8.5	12.5	12.75	18.75
115	34.5	8.0	12.0	12.0	18.0
	69	9.0	14.0	13.5	21.0
	92	10.0	15.0	15.0	23.25
138	34.5	8.5	13.0	12.75	19.5
	69	9.5	15.0	14.25	22.5
	115	10.5	17.0	15.75	25.5
161	46	9.5	15.0	13.5	21.0
	92	10.5	16.0	15.75	24.0
	138	11.5	18.0	17.25	27.0
196	46	10.0	15.0	15.0	22.5
	92	11.5	17.0	17.25	25.5
	161	12.5	19.0	18.75	28.5

(Continued)

Table D.1 (Continued)

High-voltage winding insulation class kV	Low-voltage winding insulation class kV	Impedance limits in percent			
		Class OA OW OA/FA ^{a)} OA/FA/FOA ^{a)}		Class FOA FOW	
		Min	Max	Min	Max
230	46	11.0	16.0	16.5	24.0
	92	12.5	18.0	18.75	27.0
	161	14.0	20.0	21.0	30.0

Definition of transformer classes:

- OA – Oil-immersed, self-cooled.
- OW – Oil-immersed, water-cooled.
- OA/FA – Oil-immersed, self-cooled/forced-air-cooled/forced-oil-cooled.
- FOA – Oil-immersed, forced-oil-cooled with forced-air cooler.
- FOW – Oil-immersed, forced-oil-cooled with water cooler.

Note: The through impedance of a two-winding autotransformer can be estimated knowing rated circuit voltages, by multiplying impedance obtained from this table by the factor (HV – LV/HV).

a) The impedances are expressed in percent on the self-cooled rating of OA/FA and OA/FA/FOA.

Source: Westinghouse Electric Corp. Used with permission.

Table D.2 Approximate distribution transformer impedances.

kVA	2400/4160 Y to 120/240 V 60 Hz			4800/8320 Y to 120/240 V 60 Hz			7200/12470 Y to 120/240 V 60 Hz			14440/24940 GrdY-120/240 V 60 Hz			34500 GrdY/ 19920 ^{a)} to 120/240 V 60 Hz		
	%IR	%IX	%IZ	%IR	%IX	%IZ	%IR	%IX	%IZ	%IR	%IX	%IZ	%IR	%IX	%IZ
5	1.9	1.6	2.5	2.2	1.6	2.7	2.2	2.2	3.1	2.5	2.2	3.3			
10	4.4	1.0	1.7	1.4	1.0	1.7	1.4	1.0	1.7	1.6	1.0	1.9	1.4	1.0	1.7
15	1.2	1.2	1.7	1.2	1.2	1.7	1.3	1.2	1.8	1.4	1.7	2.2	1.4	1.7	2.2
25	1.1	1.3	1.7	1.1	1.3	1.8	1.2	1.6	2.0	1.3	1.8	2.2	1.3	1.5	2.0
37.5	0.9	1.4	1.7	1.0	1.4	1.7	1.1	1.4	1.8	1.1	1.6	1.9	1.2	1.7	2.1
50	1.0	1.2	1.6	1.0	1.2	1.6	1.1	1.3	1.7	1.1	1.8	2.1	1.2	1.5	1.9
75	0.9	1.3	1.6	1.0	1.2	1.6	1.0	1.5	1.8	1.1	1.9	2.2	1.0	1.6	1.9
100	0.9	1.6	1.8	0.9	1.4	1.7	0.9	1.4	1.7	1.0	2.0	2.2	1.0	1.5	1.8
167	0.9	1.7	1.9	0.9	1.7	1.9	0.9	1.7	1.9	0.9	2.1	2.3	0.9	1.8	2.0
	240	480		240	480		240	480		240	480		240	480	
250	0.8	2.9	3.0	0.8	2.9	3.0	0.8	2.9	3.0	0.8	2.9	3.0	0.8	2.9	3.0
333	0.8	3.2	3.3	0.8	3.2	3.3	0.8	3.2	3.3	0.8	3.1	3.2	0.8	3.1	3.2
500	0.7	3.2	3.3	0.7	3.2	3.3	0.7	3.2	3.5	0.7	3.3	3.4	0.7	3.3	3.4

When only one of the two low voltage windings is loaded, the %IR and %IX (on kVA base equal to capacity of that winding, or one-half nameplate kVA) are approximately 0.75 and from 0.6 to 1.25 of full winding %IR and full winding %IX, respectively.

a) 125 kV (From General Electric Company. Used with permission.)

Table D.3 Full load currents for distribution transformer.

kVA	Single-phase circuits												
	Circuit voltage												
	120	240	480	2400	4160	4330	4880	6600	6900	7200	7620	11.5 k	13.2 k
1.5	12.5	6.3	3.1	0.63	0.36	0.35	0.31	0.23	0.22	0.21	0.20	0.13	0.11
2.5	20.8	10.4	5.2	1.04	0.60	0.58	0.52	0.38	0.36	0.35	0.33	0.22	0.19
3	25.0	12.5	6.3	1.25	0.72	0.69	0.63	0.45	0.43	0.42	0.39	0.26	0.23
5	41.7	20.8	10.4	2.08	1.20	1.16	1.04	0.76	0.72	0.69	0.66	0.43	0.38
7.5	62.5	31.3	15.6	3.13	1.80	1.73	1.56	1.14	1.09	1.04	0.98	0.65	0.57
10	83.3	41.7	20.8	4.17	2.40	2.31	2.08	1.52	1.45	1.39	1.31	0.87	0.76
15	125	62.5	31.3	6.25	3.61	3.46	3.13	2.27	2.17	2.08	1.97	1.30	1.14
25	208	104	52.1	10.4	6.01	5.77	5.21	3.79	3.62	3.47	3.28	2.17	1.89
37.5	313	156	78.1	15.6	9.01	8.66	7.81	5.68	5.43	5.21	4.92	3.26	2.84
50	417	208	104	20.8	12.00	11.55	10.4	7.58	7.25	6.94	6.56	4.35	3.79
75	625	313	156	31.3	18.00	17.32	15.6	11.4	10.9	10.4	9.84	6.52	5.68
100	833	417	208	41.7	24.00	23.10	20.8	15.2	14.5	13.9	13.1	8.70	7.58
150	1250	625	313	62.5	36.10	34.64	31.3	22.7	21.7	20.8	19.7	13.0	11.4
200	1667	833	417	83.3	48.10	46.19	41.7	30.3	29.0	27.8	26.2	17.4	15.2
250	2083	1042	521	104	60.10	57.74	52.1	37.9	36.3	34.7	32.8	21.7	18.9
333	2775	1388	694	139	80.00	76.91	69.4	50.5	48.3	46.2	43.7	29.0	25.2
500	4167	2083	1042	208	120.	115.47	104	75.8	72.5	69.4	65.6	43.5	37.9

Table D.4 Full load currents for distribution transformer.

Three-phase circuits													
kVA	Circuit voltage												
	208	240	480	2400	4160	4330	4800	6900	7200	8320	11.5 k	13.2 k	33 k
4.5	12.5	10.8	5.41	1.08	0.62	0.60	0.54	0.36	0.38	0.31	0.23	0.20	0.08
7.5	20.8	18.0	9.02	1.80	1.04	1.00	0.90	0.61	0.63	0.52	0.38	0.33	0.13
9	25.0	21.7	10.8	2.17	1.25	1.20	1.08	0.73	0.75	0.62	0.45	0.39	0.16
10	27.8	24.1	12.0	2.41	1.39	1.33	1.20	0.80	0.84	0.69	0.50	0.44	0.17
15	41.6	36.1	18.0	3.61	2.08	2.00	1.80	1.20	1.26	1.04	0.75	0.66	0.26
22.5	62.5	54.1	27.1	5.41	3.12	3.00	2.71	1.80	1.88	1.56	1.13	0.98	0.39
25	69.4	60.1	30.1	6.01	3.47	3.33	3.01	2.00	2.09	1.73	1.26	1.09	0.44
30	83.3	72.2	36.1	7.22	4.16	4.00	3.61	2.41	2.51	2.08	1.51	1.31	0.52
37.5	104	90.2	45.1	9.02	5.20	5.00	4.51	3.01	3.14	2.60	1.88	1.64	0.66
45	125	108	54.1	10.8	6.25	6.00	5.41	3.60	3.77	3.12	2.26	1.97	0.79
50	139	120	60.1	12.0	6.94	6.67	6.01	4.01	4.18	3.47	2.51	2.19	0.87
75	208	180	90.2	18.0	10.4	10.00	9.02	6.01	6.28	5.21	3.77	3.28	1.31
100	278	241	120	24.1	13.9	13.33	12.0	8.02	8.37	6.94	5.02	4.37	1.75
112.5	312	271	135	27.1	15.6	15.00	13.5	9.02	9.41	7.81	5.65	4.92	1.97
150	416	361	180	36.1	20.8	20.00	18.0	12.0	12.6	10.4	7.53	6.56	2.62
200	555	481	241	48.1	27.8	26.67	24.1	16.0	16.7	13.9	10.0	8.75	3.50
225	625	541	271	54.1	31.2	30.00	27.1	18.0	18.8	15.6	11.3	9.84	3.94
300	833	722	361	72.2	41.6	40.00	36.1	24.1	25.1	20.8	15.1	13.1	5.25
450	1249	1083	541	108	62.5	60.00	54.1	36.0	37.7	31.2	22.6	19.7	7.87
500	1388	1203	601	120	69.3	66.67	60.1	40.1	41.8	34.7	25.1	21.9	8.74
600	1665	1443	722	144	83.3	80.00	72.2	48.1	50.2	41.6	30.1	26.2	10.5
750	2082	1804	902	180	104	100.01	90.2	60.1	62.8	52.0	37.7	32.8	13.1
1000	2776	2406	1203	241	139	133.34	120	80.0	83.7	69.4	50.2	43.7	17.5
1500	4164	3609	1804	361	208	200.01	180	120	126	104	75.3	65.6	26.2